

LAB EXPERIMENT

ITA0506-COMPUTER VISION

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EXPNO:11

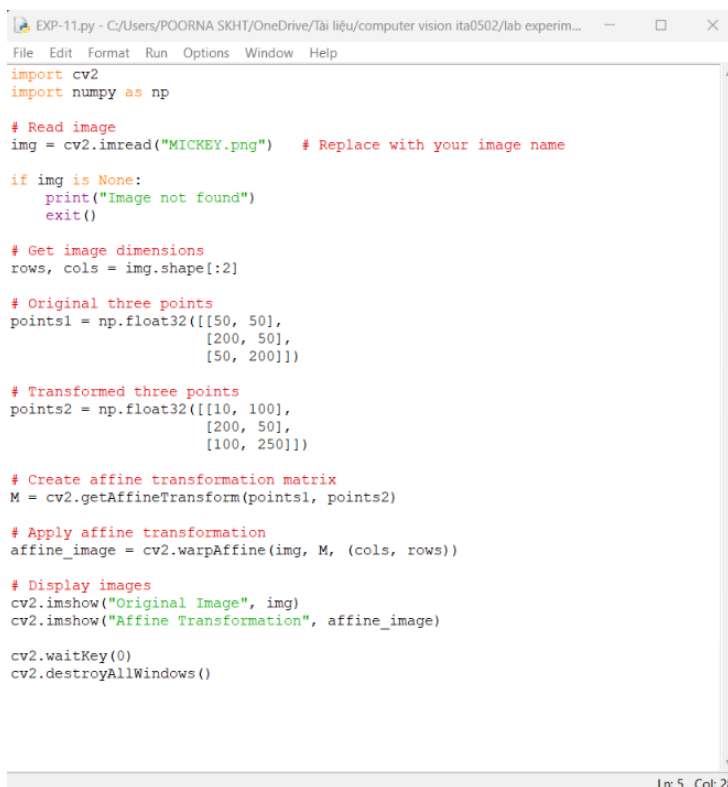
Aim:

To perform affine transformation on an image using Python and OpenCV.

Algorithm:

1. Import required libraries (OpenCV and NumPy).
2. Read the input image.
3. Select three points from the original image.
4. Define corresponding three points for the transformed image.
5. Create an affine transformation matrix using `cv2.getAffineTransform()`.
6. Apply affine transformation using `cv2.warpAffine()`.
7. Display the original and transformed images.

Code:



```
EXP-11.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...
File Edit Format Run Options Window Help
import cv2
import numpy as np

# Read image
img = cv2.imread("MICKEY.png") # Replace with your image name

if img is None:
    print("Image not found")
    exit()

# Get image dimensions
rows, cols = img.shape[:2]

# Original three points
points1 = np.float32([[50, 50],
                      [200, 50],
                      [50, 200]])

# Transformed three points
points2 = np.float32([[10, 100],
                      [200, 50],
                      [100, 250]])

# Create affine transformation matrix
M = cv2.getAffineTransform(points1, points2)

# Apply affine transformation
affine_image = cv2.warpAffine(img, M, (cols, rows))

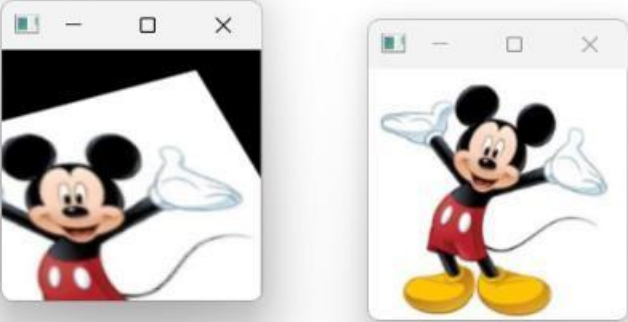
# Display images
cv2.imshow("Original Image", img)
cv2.imshow("Affine Transformation", affine_image)

cv2.waitKey(0)
cv2.destroyAllWindows()

Ln: 5 Col: 28
```

OUTPUT :

```
File Edit Shell Debug Options Window Help
Python 3.13.3 (tags/v3.13.3:6280bb5, Apr 8 2025, 14:47:33) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experiments/EXP-11.py
```



The image displays a Python 3.13.3 shell window with a menu bar (File, Edit, Shell, Debug, Options, Window, Help) and a status bar (Ln: 5 Col: 0). The shell shows a prompt, a restart command for a file named EXP-11.py, and two image windows. The left window shows a Mickey Mouse image that is tilted and partially cut off. The right window shows the same Mickey Mouse image after an affine transformation, which is now upright and fully visible within the window frame.

RESULT: The affine transformation was successfully performed on the image using OpenCV.